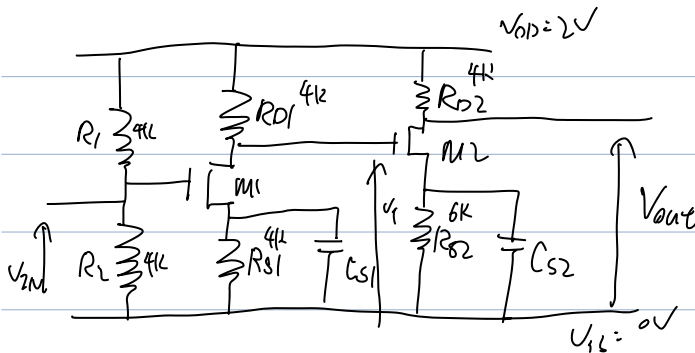


Tut 1

3.



$$\beta = 0.02 \text{ A/V}^2 \quad V_t = 0.4 \text{ V}$$

$$V_A = 10 \text{ V}$$

a. CS amplifier

Assume ideal transistor, saturation

$$I_{D1} = \frac{1}{2} \beta (V_{G1} - V_{S1} - V_t)^2$$

$$V_{G1} = \frac{R_2}{R_1 + R_2} V_{DD} = 1 \text{ V}$$

$$V_{S1} = I_{D1} \cdot R_{S1}$$

$$\Rightarrow I_{D1} = \frac{1}{2} \beta (V_{G1} - V_t - R_{S1} \cdot I_{D1})^2$$

$$\Rightarrow I_{D1} = \underline{122.35 \text{ } \mu\text{A}} \text{ or } 183.90 \text{ } \mu\text{A}$$

Threshold check: $V_{GS1} > V_t$

$$V_{GS1} = V_{G1} - I_{D1} \cdot R_{S1}$$

if $I_{D1} = 183.90 \text{ } \mu\text{A}$, then $V_{GS1} = 0.264 \text{ V} < V_t$

if $I_{D1} = 122.35 \text{ } \mu\text{A}$, then $V_{GS1} = 0.511 \text{ V} > V_t$

So I_{D1} cannot be $122.35 \text{ } \mu\text{A}$

Saturation check: $V_{DS1} > V_{GS1} - V_t$

$$V_{DS1} = V_{DD} - I_{D1}(R_{D1} + R_{S1}) = 1.02 \text{ V} > 0.511 - 0.4 = 0.111 \text{ V} \Rightarrow \text{saturation}$$

$$\Rightarrow I_{D1} = 122.35 \text{ } \mu\text{A}$$

$$i.e. \quad g_{m1} = \sqrt{2\beta_1 I_{D1}} = \sqrt{2 \times 0.02 \times 122.35 \mu} = 2.212 \text{ mA/V}$$

$$r_{o1} = \frac{V_A}{I_{D1}} = \frac{10V}{122.35 \mu} = 81.733 \text{ k}\Omega$$

b. CS amplifier

$$i.e. \quad I_{D2} = \frac{1}{2} \beta_2 (V_{G2} - V_{S2} - V_{T2})^2$$

$$V_{G2} = V_{DD} - I_{D1} \cdot R_{D1} = 2 - 122.35 \mu \cdot 4k = 1.5106 \text{ V}$$

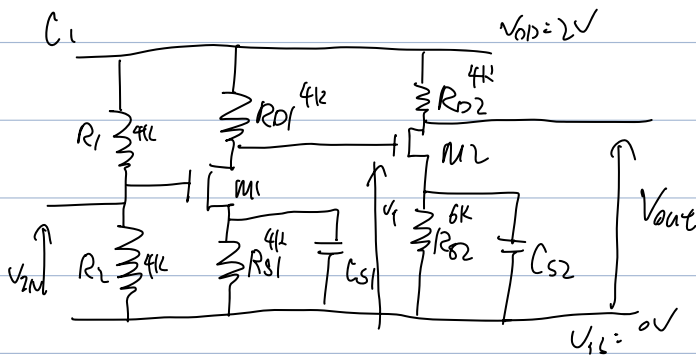
$$V_{S2} = I_{D2} \cdot R_{S2}$$

$$\Rightarrow I_{D2} = \frac{1}{2} \beta_2 (V_{G2} - V_{T2} - R_{S2} I_{D2})^2$$

$$\Rightarrow I_{D2} = \underline{163.77 \mu\text{A}} \quad \text{or} \quad 209.21 \mu\text{A}$$

$$i.e. \quad g_{m2} = \sqrt{2\beta_2 I_{D2}} = \sqrt{2 \times 0.02 \times 163.77 \mu} = 2.56 \text{ mA/V}$$

$$r_{o2} = \frac{V_A}{I_{D2}} = \frac{10V}{163.77 \mu\text{A}} = 61.061 \text{ k}\Omega$$



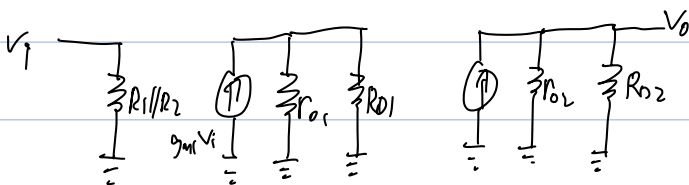
Assume saturation

$AC \ll \omega$

V_{DD} short for AC

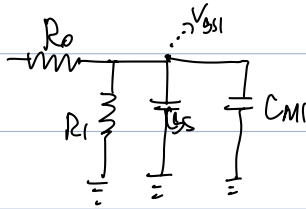
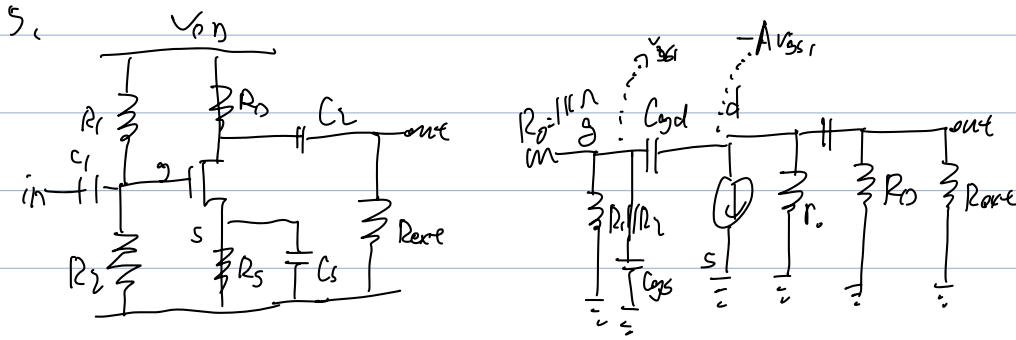
Mid-band frequency

$$G_{M1} = \frac{-g_{m1} (r_{o1} \parallel R_{D1})}{V_{in}} = g_{m1} (r_{o1} \parallel R_{D1})$$



$$G_{M2} = g_{m2} (r_{o2} \parallel R_{D2})$$

$$G_{V2} = G_{M1} G_{M2} = g_{m1} g_{m2} (r_{o1} \parallel R_{D1}) (r_{o2} \parallel R_{D2})$$



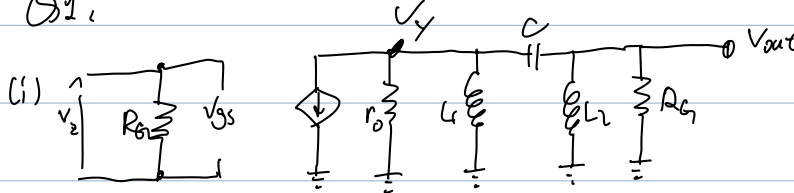
$$\frac{V_{gs1} + A V_{gs1}}{Z_{gd}} = \frac{V_{gs1}}{Z_{in1}} = \frac{0 + A V_{gs1}}{Z_{in1}}$$

$$\Rightarrow Z_{in1} = (1 + A) Z_{gd}$$

$$Z_{in1} = (1 + \frac{1}{A}) Z_{gd}$$

T_2

Q1.



Assum: AC $\ll$ DC

V_{DD} ground for AC

Mid-band frequency.

(i) $AB \gg 1 + \alpha$

$|AB| \geq 1$ and phase = $n(2\pi)$, $n = 0, 1, 2, \dots$

(ii) Ignore r_o (assume $r_o \rightarrow \infty$)

$$AB = \frac{V_{out}}{V_z} = -\frac{g_m V_z}{V_z} \cdot [Z_L \parallel (Z_C + Z_L \parallel R_G)] \cdot \frac{Z_L \parallel R_G}{Z_C + Z_L \parallel R_G}$$

$$= -g_m \cdot \frac{Z_L \cdot (Z_C + Z_L \parallel R_G)}{Z_L + Z_C + Z_L \parallel R_G} \cdot \frac{Z_L \parallel R_G}{Z_C + Z_L \parallel R_G}$$

$$= -g_m \cdot \frac{j\omega L_1 \cdot \frac{j\omega L_2 R_G}{j\omega L_2 + R_G}}{j\omega L_1 + j\omega C + \frac{j\omega L_2 R_G}{j\omega L_2 + R_G}}$$

$$= -g_m \frac{j\omega L_1 \cdot j\omega L_2 \cdot R_G}{(j\omega L_2 + R_G) (j\omega L_1 + \frac{1}{j\omega C} + \frac{j\omega L_2 R_G}{j\omega L_2 + R_G})}$$

$$= g_m \frac{\omega^2 L_1 L_2 R_G}{(j\omega L_2 + R_G) (j\omega L_1 + \frac{1}{j\omega C}) + j\omega L_2 R_G}$$

$$= g_m \frac{\omega^2 L_1 L_2 R_G}{-\omega^2 L_1 L_2 + \frac{L_2}{C} + j\omega L_1 R_G + \frac{R_G}{j\omega C} + j\omega L_2 R_G}$$

$$= \frac{g_m \omega^2 L_1 L_2 R_G}{\frac{L_2}{C} - \omega^2 L_1 L_2 + j\omega R_G (L_1 + L_2 - \frac{1}{\omega^2 C})}$$

$$\Rightarrow \omega = \sqrt{\frac{1}{(L_1 + L_2)C}} \Rightarrow f_0 = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{L_1 + L_2}C}$$

$$\frac{g_m \frac{L_1 L_2}{(L_1 + L_2)C} \cdot R_G}{\frac{L_2}{C} - \frac{L_1 L_2}{(L_1 + L_2)C}} > 1$$

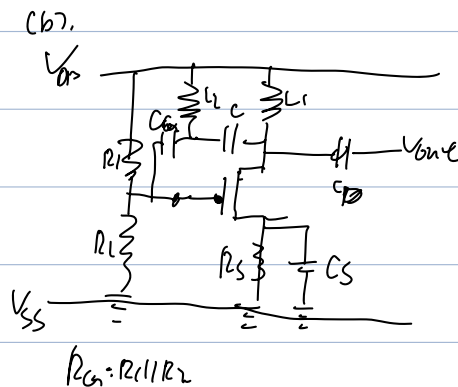
$$\Rightarrow g_m (L_1 / L_2) R_G > L_2 - L_1 / L_1$$

$$\Rightarrow g_m \geq \frac{L_1 + L_2}{L_1} \cdot \frac{1}{R_G} - \frac{1}{R_G} = \frac{L_2}{L_1 R_G}$$

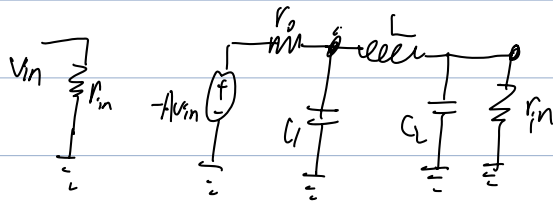
Q2. (a) Barkhausen Criteria $A_{CL} \geq 1 + j0$

$$A_{CL} = -g_m \frac{R_G}{1 - \omega^2 L C_1 + j\omega R_G (C_1 + C_2 - \omega^2 L C_1 C_2)}$$

$$\Rightarrow \omega = \sqrt{\frac{C_1 + C_2}{L C_1 C_2}} \Rightarrow f_r = \frac{1}{2\pi} \cdot \sqrt{\frac{C_1 + C_2}{L C_1 C_2}}$$



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$$A_{in} = \frac{-A_{vin}}{V_{in}} \cdot \frac{C_1 // [L + C_2 // r_{in}]}{r_o + C_1 // [L + C_2 // r_{in}]} \cdot \frac{C_2 // r_{in}}{L + C_2 // r_{in}}$$

$$= -A \cdot \frac{\cancel{C_1} // [L + C_2 // r_{in}]}{r_o + \frac{C_1(L + C_2 // r_{in})}{C_1 + L + C_2 // r_{in}}} \cdot \frac{C_2 // r_{in}}{\cancel{L + C_2 // r_{in}}}$$

$$= -A \cdot \frac{\frac{1}{j\omega C_1} \cdot \left(\frac{1}{j\omega C_2} // r_{in} \right)}{r_o \left(\frac{1}{j\omega C_1} + j\omega L + \frac{1}{j\omega C_2} // r_{in} \right) + \frac{1}{j\omega C_1} \left(j\omega L + \frac{1}{j\omega C_2} // r_{in} \right)}$$

$$= A \cdot \frac{\frac{1}{j\omega C_1} \cdot \frac{1}{j\omega C_2} \cdot r_{in}}{r_o \left[\left(\frac{1}{j\omega C_1} + j\omega L \right) \left(\frac{1}{j\omega C_2} + r_{in} \right) + \frac{1}{j\omega C_2} \cdot r_{in} \right] + \frac{L}{C_1} \left(\frac{1}{j\omega C_2} + r_{in} \right) + \frac{1}{j\omega C_1} \left(\frac{r_{in}}{j\omega C_2} \right)}$$

$$= \frac{\frac{A r_{in}}{\omega^2 C_1 C_2}}{r_o \left[-\frac{1}{\omega^2 C_1 C_2} + \frac{L}{C_2} + \frac{r_{in}}{j\omega C_1} + j\omega L r_{in} + \frac{r_{in}}{j\omega C_2} \right] + \frac{L}{j\omega C_1 C_2} + \frac{L r_{in}}{C_1} - \frac{r_{in}}{\omega^2 C_1 C_2}}$$

$$= \frac{\frac{A r_{in}}{\omega^2 C_1 C_2}}{r_o \cdot \left[-\frac{1}{\omega^2 C_1 C_2} + \frac{L}{C_2} \right] + \left[\frac{L}{C_1} - \frac{1}{\omega^2 C_1 C_2} \right] r_{in} - \frac{1}{j\omega C_1 C_2} ()}$$

$$\Rightarrow \frac{r_{in} r_o}{j\omega C_1} + j\omega L r_{in} r_o + \frac{r_{in} r_o}{j\omega C_2} + \frac{L}{j\omega C_1 C_2} = 0$$

$$\Rightarrow C_2 r_{in} r_o - \omega^2 L C_1 C_2 r_{in} r_o + C_1 r_{in} r_o + L = 0$$

$$\Rightarrow \omega^2 = \frac{r_{in} r_o (C_1 + C_2) + L}{L C_1 C_2 r_{in} r_o}$$

$$= \frac{C_1 + C_2}{L C_1 C_2} + \frac{1}{C_1 C_2 r_{in} r_o}$$

$$A_{eq} = \frac{A r_{in}}{r_o [\omega^2 L C_1 - 1] + r_{in} [\omega^2 L C_2 - 1]}$$

$$= \frac{A r_{in}}{r_o \left[\frac{C_1}{C_2} + \frac{L}{C_2 r_{in} r_o} \right] + r_{in} \left[\frac{C_1}{C_2} + \frac{L}{C_2 r_{in} r_o} \right]}$$

$$= \frac{A r_{in}}{\frac{C_1 r_o}{C_2} + \frac{L}{C_2 r_{in}} + \frac{C_1 r_{in}}{C_2} + \frac{L}{C_2 r_o}} \gg 1$$

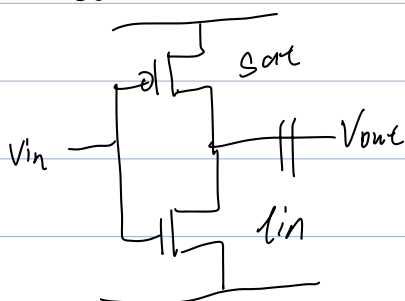
$$A \gg \frac{1}{r_{in}} \left[\frac{C_1 r_o}{C_2} + \frac{C_1 r_{in}}{C_2} + \frac{L}{C_2 r_{in}} + \frac{L}{C_2 r_o} \right]$$

$$\approx \frac{1}{r_{in} C_2} \left[C_1^2 r_o + C_1^2 r_{in} + \frac{C_1 L}{r_{in}} + \frac{C_1 L}{r_o} \right]$$

$$\frac{1}{1k} \left[1k + 1k + \frac{100n}{1k} + \frac{100n}{1k} \right]$$

$$\approx 2$$

T3. Q2.



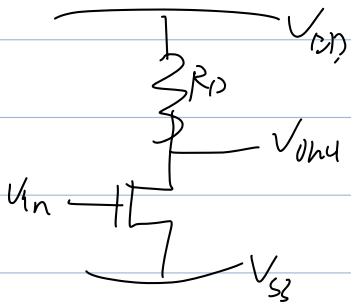
$$\frac{\beta_p}{2} (V_{DD} - V_{in} - V_{tp})^2$$

$$\frac{\beta_n}{2} [2(V_{in} - V_{tn})V_{out} + V_{out}^2]$$

$$V_{out}^2 - 2(V_{in} - V_{tn})V_{out} + \frac{\beta_p}{\beta_n} (V_{DD} - V_{in} - V_{tp})^2 = 0$$

$$\Rightarrow V_{out} = (V_{in} - V_{tn}) \pm \sqrt{(V_{in} - V_{tn})^2 - \frac{\beta_n}{\beta_p} (\dots)^2}$$

$$\frac{\beta_p}{2} (V_{DD} - V_{in} - V_{tp})^2 = \frac{\beta_n}{2} (V_{in} - V_{tn})^2$$



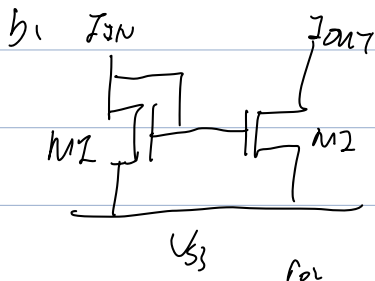
T4. Q1, Assume M_1, M_2 are identical

$$I_{DN} = \frac{\beta_1}{2} (V_G - V_S - V_{t1})^2$$

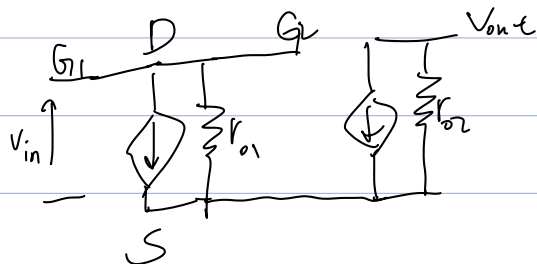
$$I_{DN} = \frac{\beta_2}{2} (V_G - V_S - V_{t2})^2$$

$$\Rightarrow I_{DN} = \frac{\beta_1}{\beta_2} I_{DN} \quad \beta = \frac{\mu W}{L}$$

$$= I_{DN}$$



$\Rightarrow$



$$c. \quad \frac{I_{DN}}{I_{DN}} = 2$$

Q2. Assume ideal, sat

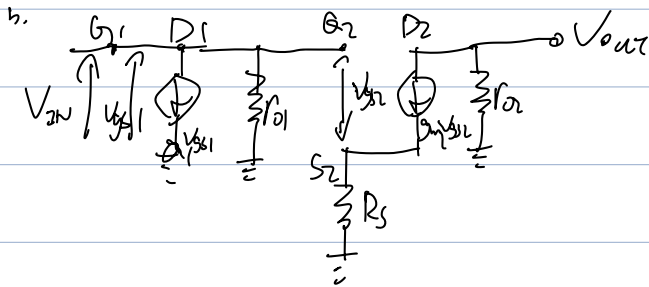
$$I_{D1} = \frac{\beta_1}{2} (V_{G1} - V_{S1} - V_{t1})^2$$

$$I_{D2} = \frac{\beta_2}{2} (V_{G2} - V_{S2} - V_{t2})^2 = \frac{\beta_2}{2} (V_{G1} - I_{D1} R_S - V_{t2})^2$$

$$\Rightarrow \sqrt{\frac{2I_{D1}}{\beta_1}} = V_{G1} - V_{S1} - V_{t1} \Rightarrow V_{G1} = \sqrt{\frac{2I_{D1}}{\beta_1}} + V_{t1}$$

$$\Rightarrow \sqrt{\frac{2I_{D2}}{\beta_2}} = V_{G1} - I_{D1} R_S - V_{t2} \Rightarrow V_{G1} = \sqrt{\frac{2I_{D2}}{\beta_2}} + I_{D1} R_S + V_{t2}$$

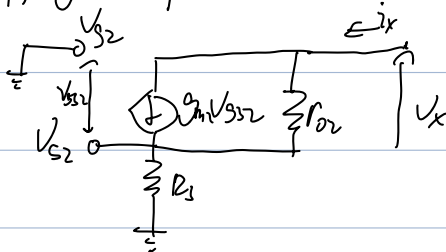
$$\Rightarrow \sqrt{\frac{2I_{D1}}{\beta_1}} - \sqrt{\frac{2I_{D2}}{\beta_2}} = I_{D1} R_S$$



Assume mid-band, AC << f_c

superposition

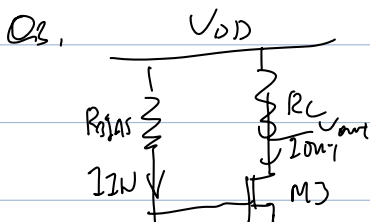
Apply inspection node:



$$\begin{cases} i_k - g_m(V_{S2}) = \frac{V_x - V_{S2}}{R_{D2}} \\ V_{S2} = i_k R_S \end{cases}$$

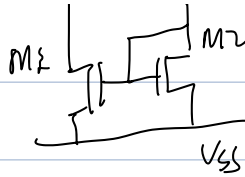
$$\Rightarrow i_k + g_m R_S i_k = \frac{V_x}{R_{D2}} - \frac{R_S}{R_{D2}} i_k$$

$$\Rightarrow R_{out} = \frac{V_x}{i_k} = R_{D2} \left[1 + g_m R_S + \frac{R_S}{R_{D2}} \right] \\ = R_{D2} (1 + g_m R_S) + R_S$$



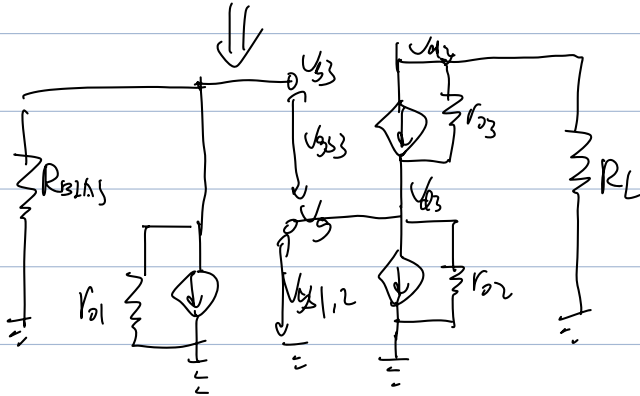
$$I_1 = \frac{\beta_1}{2} (V_G - V_{t1})^2 (1 + \lambda V_{D1})$$

$$I_2 = \frac{\beta_2}{2} (V_G - V_{t2})^2 (1 + \lambda V_{D2})$$



$$\frac{I_{out}}{I_{in}} = \frac{\beta_2}{\beta_1} \frac{1 + \lambda V_{gs}}{1 + \lambda V_{ds2}}$$

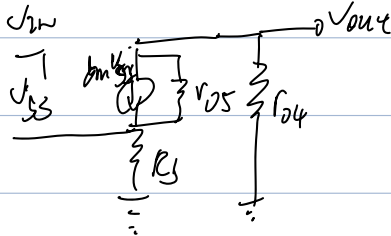
$$V_{ds2} \approx 2V_{gs}$$



Q4. $10M N: \beta = 0.01 A/V^2$

$10M P: \beta = 0.004 A/V^2$ $V_A = 10V$ $I_{DN} = 150 \mu A$

$$g_{m1} = \sqrt{2\beta_N I_{DN}} = \sqrt{2 \times 0.01 \times 150 \mu A}$$



$$\frac{V_{out}}{r_{04}} + g_m V_{gs5} + \frac{V_{out} - V_{gs5}}{r_{05}} = 0$$

$$-\frac{V_{out}}{r_{04}} = \frac{V_{gs5}}{R_S} \Rightarrow V_{gs5} = -\frac{R_S}{r_{04}} V_{out}$$

$$\Rightarrow \frac{V_{out}}{r_{04}} + g_m \left(V_{gs1} + \frac{R_S}{r_{04}} V_{out} \right) + \frac{V_{out} + \frac{R_S}{r_{04}} V_{out}}{r_{05}} = 0$$

$$\Rightarrow \frac{A}{r_{04}} + g_m \left(1 + \frac{R_S}{r_{04}} A \right) + \frac{A}{r_{05}} + \frac{R_S}{r_{04} r_{05}} A = 0$$

$$\Rightarrow A = \frac{-g_m}{\frac{1}{r_{04}} + \frac{R_S}{r_{04}} g_m + \frac{1}{r_{05}} + \frac{R_S}{r_{04} r_{05}}}$$

$$= \frac{-g_m r_{04} r_{05}}{r_{05} + R_S r_{05} g_m + r_{04} + R_S}$$

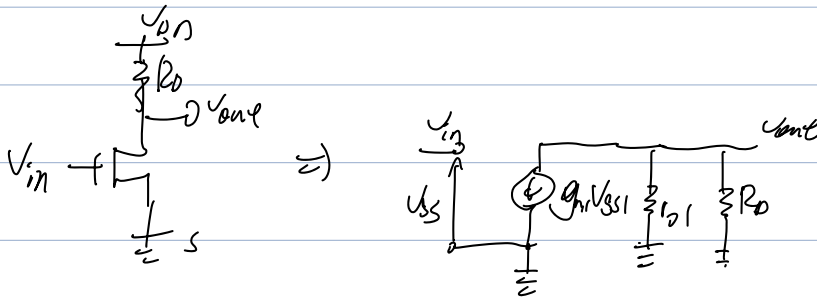
$$Q_{D1}, a_1 I_1 = \frac{\beta_1}{2} \left(\frac{V_{DD}}{2} - V_S - V_{th} \right)^2 = I_L$$

$$V_S = (I_1 + I_L) R_S = 2 I_1 R_S$$

$$\Rightarrow I_1 = \frac{\beta_1}{2} \left(\frac{V_{DD}}{2} - 2 I_1 R_S \right)^2$$

$$\Rightarrow \pm \sqrt{\frac{2 I_1}{\beta_1}} = \left(\frac{V_{DD}}{2} - 2 R_S I_1 \right) \Rightarrow I_1 = \frac{104 \mu A}{77.5 \mu A}$$

$$g_m = \frac{V_{out}}{V_{in}}$$



$$r_{o1} = \frac{V_A}{I_D} = 129 \text{ k}\Omega \gg R_D \approx 5 \text{ k}\Omega$$